

EUGENE L. BUCHANAN

Staff AI Engineer — Multi-Agent Frameworks & Agentic Infrastructure

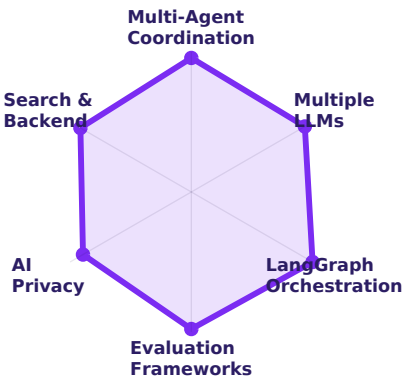
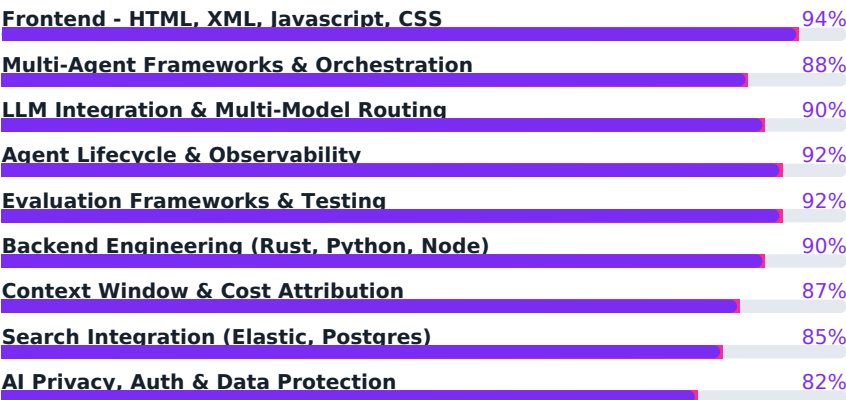


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Professional Summary

Using probabilistic systems to create deterministic solutions reliably. That is the core of my work: designing and shipping agentic AI infrastructure where teams build, deploy, and coordinate intelligent agents at scale. I turn ambiguous model behavior into predictable, observable, well-bounded systems teams can trust in production — multi-agent orchestration, multi-model routing, evaluation, and the privacy and search foundations that make AI-native platforms real.

Technical Expertise



Technology Stack

- LangGraph Multi-Agent Coordination Agent Lifecycle JSON-RPC Bounded I/O Fire-and-Forget Unix Domain Sockets OpenCode
- Codex Agents ZeroCode Hermes Agent Claude Code
- Context Window Routing Multi-Model Cost Attribution MCP (Model Context Protocol) External Tool Access Prompt Engineering RAG
- OpenAI Anthropic Gemini Ollama Hugging Face
- Rust Python Node.js PostgreSQL AWS (ECS, S3) REST GraphQL Docker Git Slack Jira Bitbucket Confluence
- Cypress Selenium JUnit Postman Elasticsearch Regression Suites Health Dashboards Alerting Qase
- SAML OAuth OIDC Permission Profiles Principal Isolation Deny-by-Default SOC 2 SCORM LTI Webhooks

Key Strengths

Key Strengths — select a language:

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Translations:

Multi-agent orchestration — design and ship multi-agent systems in production with lifecycle observability.
Multi-model routing & cost — unify LLM providers behind one context window with per-model cost attribution.
Evaluation frameworks — build harnesses, regression suites, health dashboards, and alerting.
LLM integration — prompt engineering and chatbot tooling in shipped products and automation.
Cross-functional leadership — lead as Product Owner, QA Lead, and presales engineer.

Experience

Staff AI Engineer / Agentic Infrastructure (Open-Source)

2026 – Present

ZeroClaw Labs (github.com/zeroclaw-labs/zeroclaw) · Rust · 32.8k stars

- PR #7946 (MERGED Jul 2026) — context_window source of truth for 9 providers (OpenRouter + 8 OpenAI-compatible); doctor command, gateway API, ContextBar. Reviewed by WareWolf-MoonWall, Audacity88, singlerider, JordanTheJet.
- PR #8966 (OPEN) — live provider identity on usage events (provider_ref, model, accepted); per-provider ledger on done frame; trim budget split from model capacity. Fixes stream-failure fallback #10736.
- PR #8337 (CLOSED, superseded by #10269) — lifecycle observer via JSON-RPC over UDS. Four review rounds cleared (bounded I/O, env discovery, scope gating, terminal drain).

Senior QA Engineer / Test Automation Lead / Integrations Product Owner

Dec 2020 – Aug 2025

Knowledgecity LLC · Remote · AWS / Postgres / React

- Product Owner, Integrations: shipped SAP, Oracle, Workday, Coursera, UKG, and Zoom integrations via SAML, OAuth, SFTP, REST APIs, webhooks, SCORM, and LTI — one surface across auth models and data formats.
- Built and maintained CI/CD quality pipelines with Cypress, Selenium, Postman, JUnit, Qase, Slack, Jira, Bitbucket, Confluence, Elastic, S3 — evaluation frameworks for complex deployment scenarios.
- Helped develop AI support chatbot with RAG retrieval + guardrails (SOC 2); shipped LLM-powered tooling into customer-facing assistance and internal automation workflows.
- Designed evaluation frameworks, regression suites, health dashboards, and alerting across integration, backend, API, database, and frontend surfaces; joined blue-team incident forensics.
- Mentored and trained the test automation team; ran code reviews; raised engineering standards.

Senior Software Engineer II / Presales

1999 – 2001

RealNetworks · Seattle, WA · Streaming Media Platform

- SDET QA across Linux/Unix/Windows/Embedded for the streaming media platform; advanced research team spanning appliances, mobile, stream servers, cellular networks (TFRCP patent work).
- Global presales engineering with VP sales support and travel to Japan and Korea. Trained new hires.

Senior Software Test Engineer IV

1999

Microsoft · Redmond, WA · Windows Media Server 4/5

- Nightly build test script automation; white-box testing and development for complex system testing.

Software Test Engineer II

1997

Microsoft · Redmond, WA · Windows 98/NT

- OEM setup and hardware/driver verification across Windows 98/NT releases.

Earlier Roles — Test & Helpdesk

1996 – 1998

IBM · Keene Inc. · Kirkland / Seattle, WA

- IBM Senior Test Engineer III (1998): high-availability servers, WHQL cluster-failover certification.
- Keene Inc. Technical Support Agent II (1996): frontline and escalated helpdesk ticketing, Windows 95/NT.

Education

- Associate Degree, Victor Valley College
- California Notary Commission
- Toastmasters International
- High School Diploma, Lucerne Valley High School

Other Interests

Renewable Energy

Solar, storage, and the grid of the future.

Real Estate

Buy-and-hold multifamily, cash-flow-first.

Permaculture

Food forests, rainwater harvesting, and topsoil.

Tango Dancing

Argentine tango — embrace, pause, musicality.



GitHub — eugeneb50